



Manufacturing Processes Group Project

Members of team 11:

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Mohamad Nour edeen 300450

4 x $\varnothing$ 13.50 THRU ALL
 $\varnothing$ 23.79 ∇ 7.50

R92.50

20.00

6.29

A $\varnothing$ 150.00

10°

50.00

100.00

$\varnothing$ 220.00

R27.50

75.00

20.00

$\varnothing$ 80.00

$\varnothing$ 120.00

SECTION A-A

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:				FINISH:		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING		REVISION	
NAME				SIGNATURE		DATE		TITLE			
DRAWN								MATERIAL: C45 WEIGHT: DIV'G NO. casestudy35 SCALE 1:2 SHEET 1 OF 1			
CHK'D											
APP'VD											
MFG											
Q.A.											

SOLIDWORKS Educational Product. For Instructional Use Only.

Nominal part

nominal pp

Options...

Override Mass Properties... Recalculate

☒ Include hidden bodies/components
☐ Create Center of Mass feature
☐ Show weld bead mass

Report coordinate values relative to: -- default --

Mass properties of nominal pp
Configuration: Default
Coordinate system: -- default --

Density = 0.00 grams per cubic millimeter

Mass = 2296.83 grams

Volume = 2296832.15 cubic millimeters

Surface area = 146758.17 square millimeters

Center of mass: (millimeters)
X = 0.08
Y = 16.64
Z = 0.00

Principal axes of inertia and principal moments of inertia: (grams * square millimeter)
Taken at the center of mass.
I_x = (0.00, 0.00, 1.00) P_x = 7882580.76
I_y = (1.00, 0.00, 0.00) P_y = 7887586.21
I_z = (0.00, 1.00, 0.00) P_z = 13110760.29

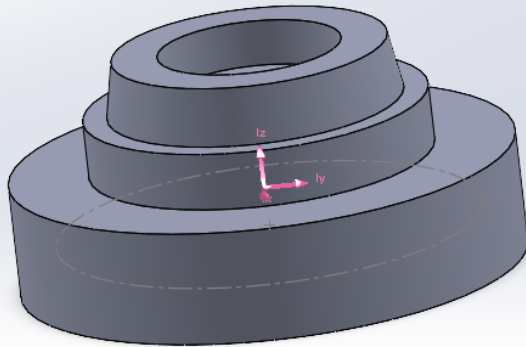
Moments of inertia: (grams * square millimeters)
Taken at the center of mass and aligned with the output coordinate system. (Using positive tensor notation.)
L_{xx} = 7887587.28 L_{xy} = 2365.83 L_{xz} = 0.00
L_{yx} = 2365.83 L_{yy} = 13110759.22 L_{yz} = 0.00
L_{zx} = 0.00 L_{zy} = 0.00 L_{zz} = 7882580.76

Moments of inertia: (grams * square millimeters)
Taken at the output coordinate system. (Using positive tensor notation.)
I_{xx} = 8523643.64 I_{xy} = 5312.95 I_{xz} = 0.00
I_{yx} = 5312.95 I_{yy} = 13110772.87 I_{yz} = 0.00
I_{zx} = 0.00 I_{zy} = 0.00 I_{zz} = 8518650.78

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Check Body Compare Deviation Analysis Draft Analysis Symmetry Check Check Active Document 3DEX
Geometry Analysis Zebra Stripes Undercut Analysis Thickness Analysis Compare Documents
Import Diagnostics Curvature Parting Line Analysis

-Ins Simulation MBD SOLIDWORKS CAM SOLIDWORKS CAM TBM Analysis Preparation



Material Type

Category of steel used

→ In particular, material density must be reported.

→ Material group is defined by EN10277-2: 2008 standard.

→ Group M1: Steel with carbon content not more than 0,65% and total of specified alloying elements (Mn, Ni, Cr, Mo, V, W) not more than 5% by mass.

→ Group M2: Steel with carbon content above 0,65% or total of specified alloying elements (Mn, Ni, Cr, Mo, V, W) above 5% by mass.

Material:	c45
Material type:	M1
Density (kg/m ³):	7514
Density (kg/mm ³):	0.000007514

Shape complexity factor of the nominal part

→ Complexity factor S is defined by EN DIN 10243-1: 1999 standard.

→ Complexity factor S is computed as mass of the nominal part / mass of the enveloping.

Vnom (mm ³):	2296832
Mnom (kg):	17.25839565

imported from solidworks

Dmax (mm):	220.00
Hmax (mm):	100.00
Venv (mm ³):	3801327.11
Menv (kg):	28.56

Complexity categories S		
0.63	< S1 <=	1
0.32	< S2 <=	0.63
0.16	< S3 <=	0.32
0	< S4 <=	0.16

S:	0.60
Category:	S2

Forged Part

forged_part_casestudy

Options...

Override Mass Properties... Recalculate

☒ Include hidden bodies/components
☐ Create Center of Mass feature
☐ Show weld bead mass

Report coordinate values relative to: -- default --

Mass properties of forged_part_casestudy
Configuration: Default
Coordinate system: -- default --

Density = 0.00 grams per cubic millimeter

Mass = 2666.53 grams

Volume = 2666534.16 cubic millimeters

Surface area = 142786.72 square millimeters

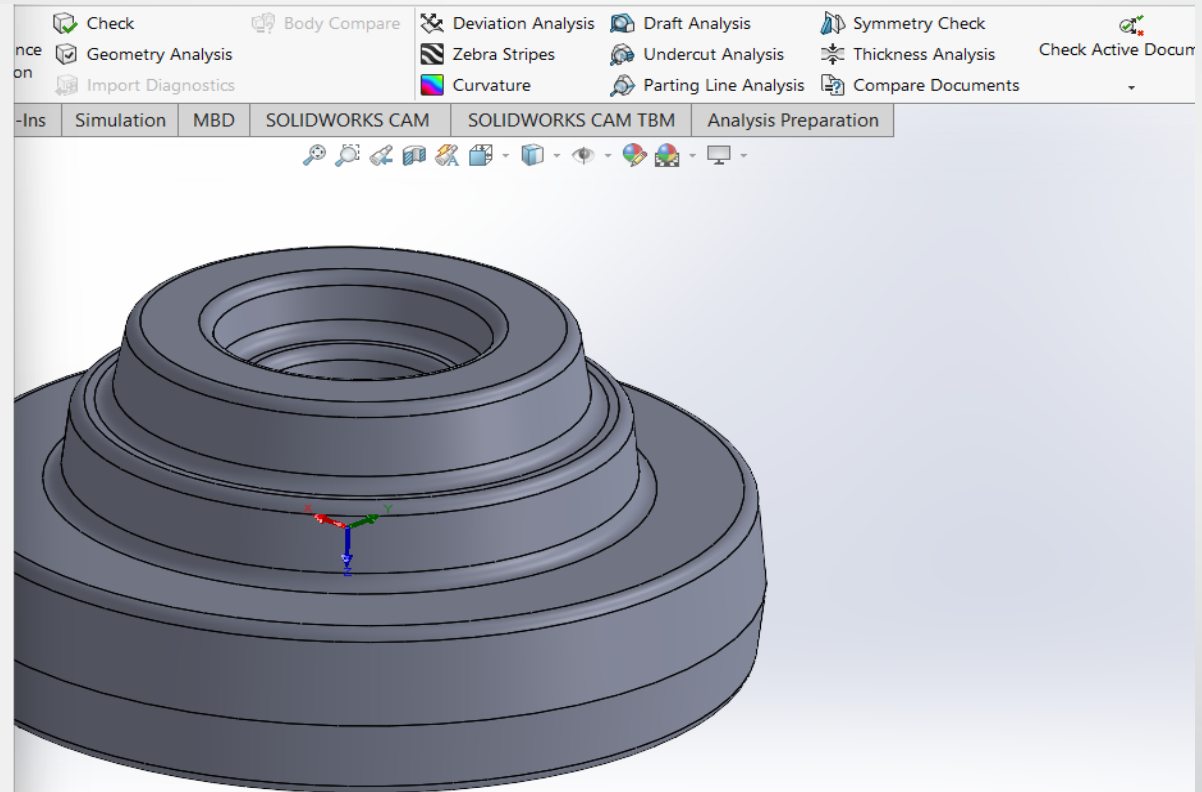
Center of mass: (millimeters)
X = 0.00
Y = 0.00
Z = -3.33

Principal axes of inertia and principal moments of inertia: (grams * square millimeter)
Taken at the center of mass.
I_x = (1.00, 0.00, 0.00) P_x = 9362298.04
I_y = (0.00, 1.00, 0.00) P_y = 9362298.04
I_z = (0.00, 0.00, 1.00) P_z = 15419636.07

Moments of inertia: (grams * square millimeters)
Taken at the center of mass and aligned with the output coordinate system. (Using positive tensor notation.)
L_{xx} = 9362298.04 L_{xy} = 0.00 L_{xz} = 0.00
L_{yx} = 0.00 L_{yy} = 9362298.04 L_{yz} = 0.00
L_{zx} = 0.00 L_{zy} = 0.00 L_{zz} = 15419636.07

Moments of inertia: (grams * square millimeters)
Taken at the output coordinate system. (Using positive tensor notation.)
I_{xx} = 9391879.68 I_{xy} = 0.00 I_{xz} = 0.00
I_{yx} = 0.00 I_{yy} = 9391879.68 I_{yz} = 0.00
I_{zx} = 0.00 I_{zy} = 0.00 I_{zz} = 15419636.07

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Dimensions and the complexity factor of the Forged Part

	Nominal part dimension (mm)	Forging grade	Tolerance (mm)	Tolerance with limits (mm)	Total forging dimension (mm)	Final dimensions with temperature scaling
Length (Table 1)	220.00	F	3.60	3.6(+2.4/-1.2)	223.60	Neglected
	150.00	F	3.20	3.2(+2.1/-1.1)	153.20	
	80.00	F	2.80	2.8(+1.9/-0.9)	77.20	
	55.00	F	2.80	2.8(+1.9/-0.9)	52.20	
	120.00	F	3.20	3.2(+2.1/-1.1)	123.20	
Thickness (Table 3)	100.00	F	3.20	3.2(+2.1/-1.1)	103.20	
Height (Table 1)	50.00	F	2.80	2.8(+1.9/-0.9)	52.80	
	20.00	F	1.50	2.5(+1.7/-0.8)	18.50	
	25.00	F	0.50	2.5(+1.7/-0.8)	25.50	

2.5 Study complexity factor of the forged part		
→ Complexity factor S is computed as mass of the forged part / mass of the enveloping.		
→ Complexity factor S may change between nominal and forged part		
Vforged (mm³):	2666534.16	from solidworks
Mforged (kg):	20.04	
Dmax (mm):	223.60	
Hmax (mm):	103.20	
Venv (mm³):	4052408.12	
Menv (kg):	30.44979463	
S:	0.66	
Category:	s1	

We can see a change from s2 in the nominal to s1 in the forged part

Flash dimensions

Flash dimensioning

- It is required to evaluate b and s for the geometry of the flash.
- The computation is developed through Bruchanow and Rebelsky formula.

Thickness s (mm):	2.972
b/s ratio:	3
Die land b (mm):	7.431
Projected area with flash (mm ²):	44660.989

Consider Enlarging

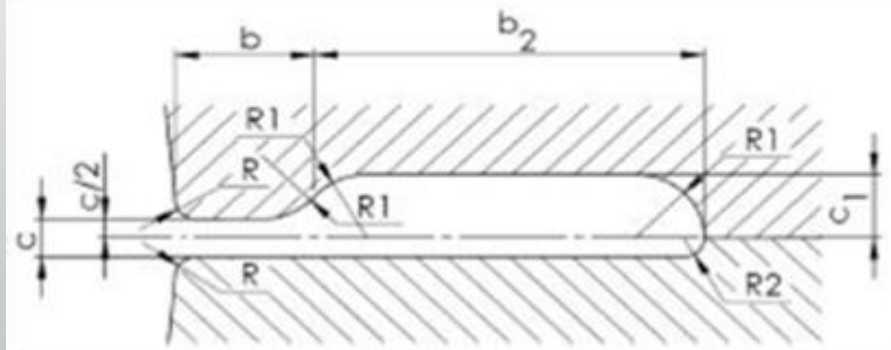
Consider radius of forged part + b

Heuristic design: Bruchanow e Rebelski

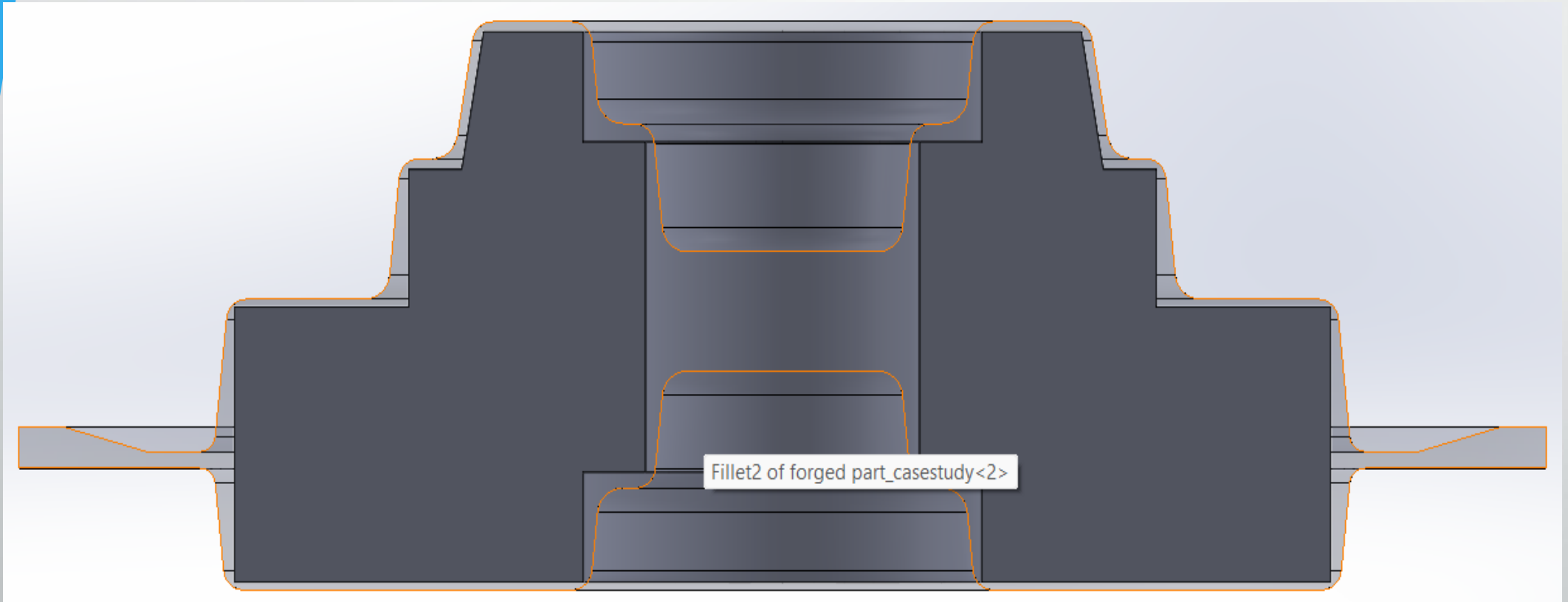
$A_{f,p}$ Projected area of the part (no flash).

$$s = 0,015 \cdot \sqrt{A_{f,p}}$$

$D_{f,p}$ [mm]	$A_{f,p}$ [mm ²]	S_{max} [mm]	b/s		
			Upsetting	Enlarging	Filling
Fino a 50	1962	0,6	8	10	13
50-80	1962-5024	1,0	7	8	10
120-190	11.200-28.000	2,5	4	4,5	5,5
300-480	71.000-180.000	6,3	2	2,5	3
480-760	180.000-450.000	10,0	1,5	2	2,5



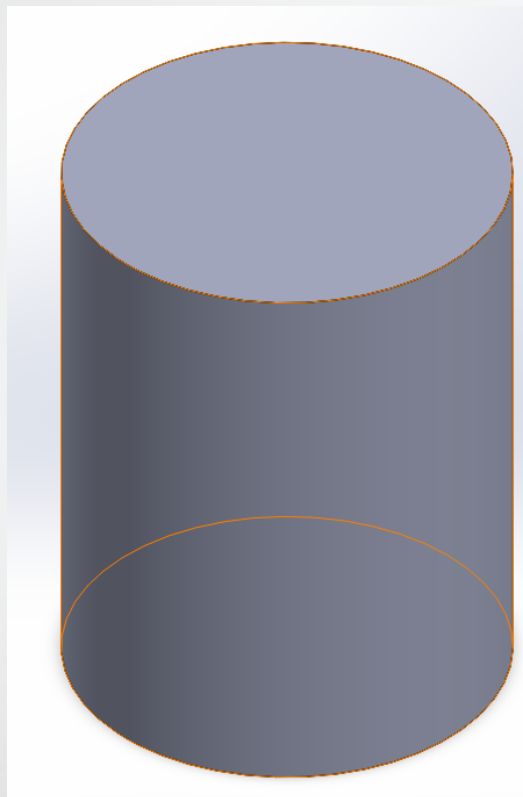
Forged Part Including the Nominal Part

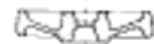


Here we check fillet if include all material

Billet Dimensioning

Material density (kg/mm ³):	0.000007514
Category of shape:	2
K:	1.08
Mass forged part * K:	20.736
Total volume:	2759648.66
Diameter Dg (mm):	120
Height Hg (mm):	244.00
h/d < 2,5	2.03



Ref. shapes	GRUPPO
	1.1
	1.2
	2.1
	2.2
	3.1
	3.2

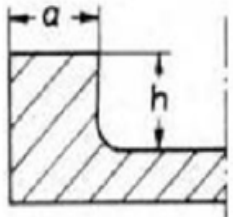
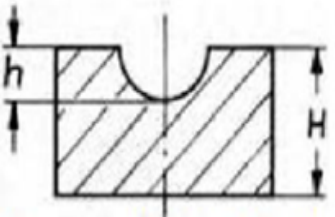
M _{nom} [Kg]/ Category of shape		1	2,5	4	6,3	20	100
K	1	1,1	1,08	1,07	1,06	1,05	1,03
	2	1,25	1,19	1,17	1,15	1,08	1,06
	3	1,5	1,46	1,41	1,35	1,2	

Die Block Dimensions

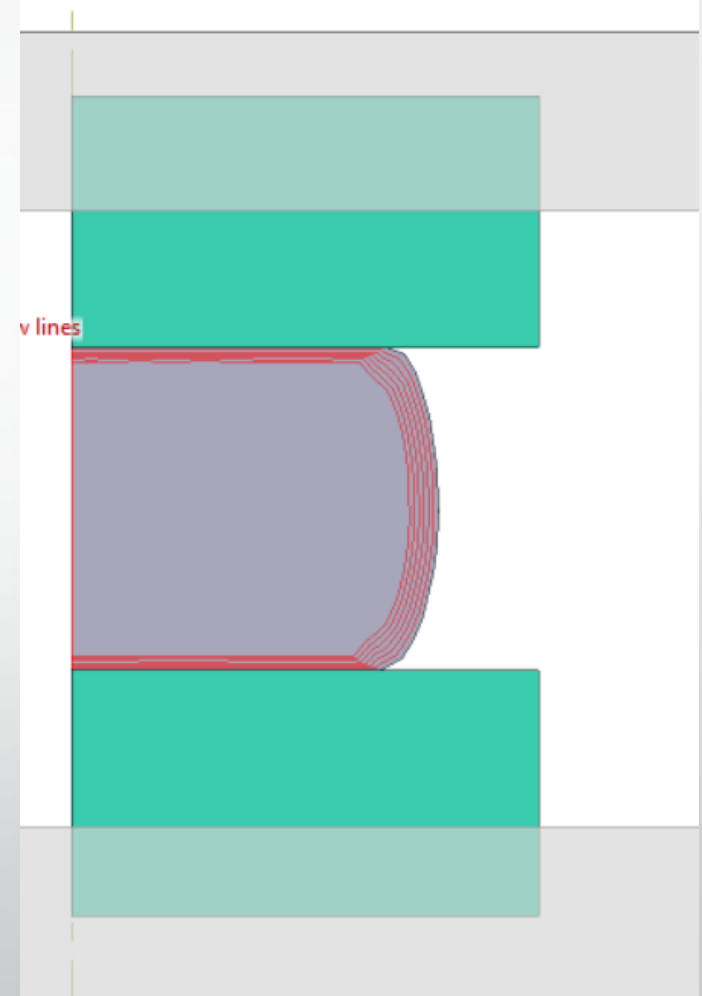
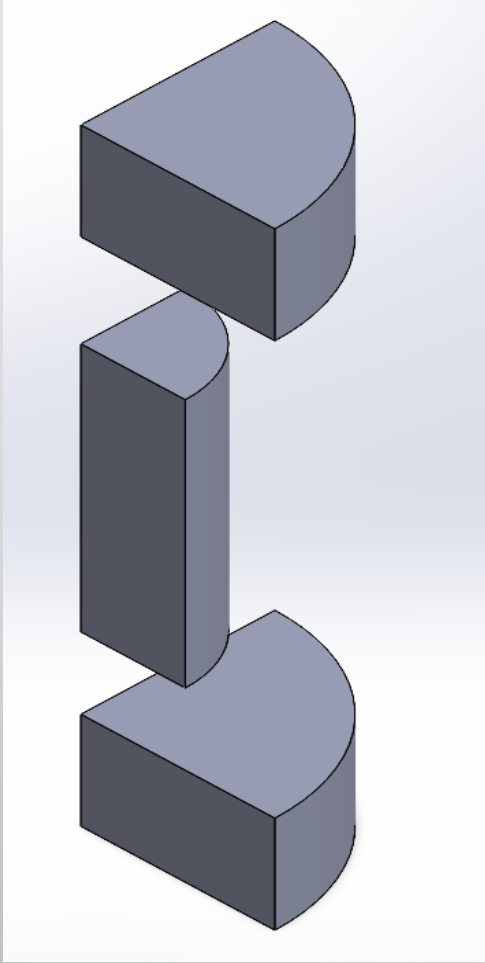
Die block dimensions

- The dies must be of an appropriate size, to accommodate for the stresses, that are achieved during the forging process.
- Required only if the coupled deformation is simulated.
- The dimensions selected for the die blocks depend on the depth of the cavity.

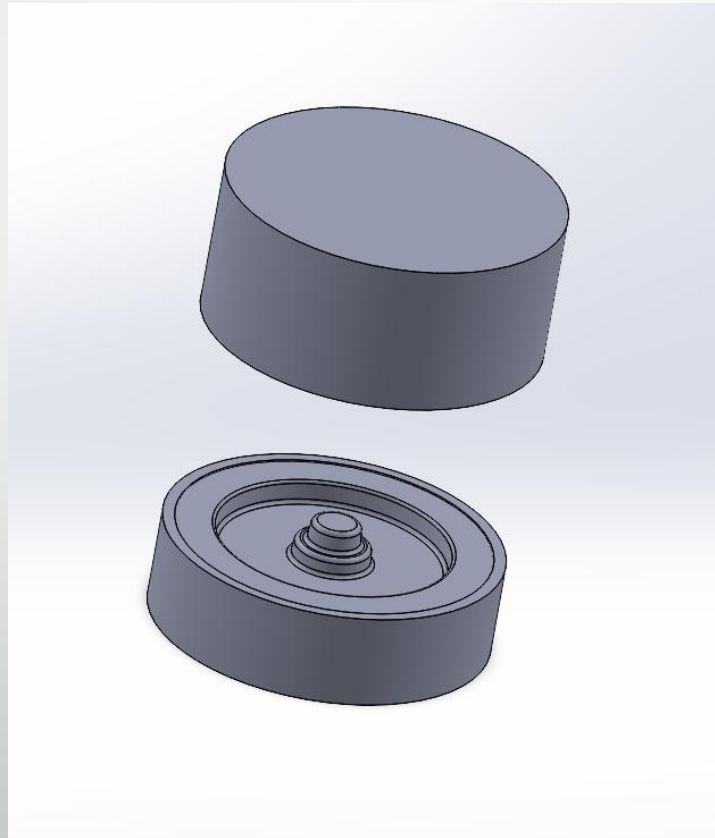
Cavity depth h (mm):	78
Min thickness (mm):	120
Min height (mm):	330

Cavity depth h [mm]		
	Minimum thickness a [mm]	Minimum die block height H [mm]
6	12	100
10	20	100
16	32	125
25	40	160
40	56	200
63	80	250
100	110	315
125	130	355
160	160	400

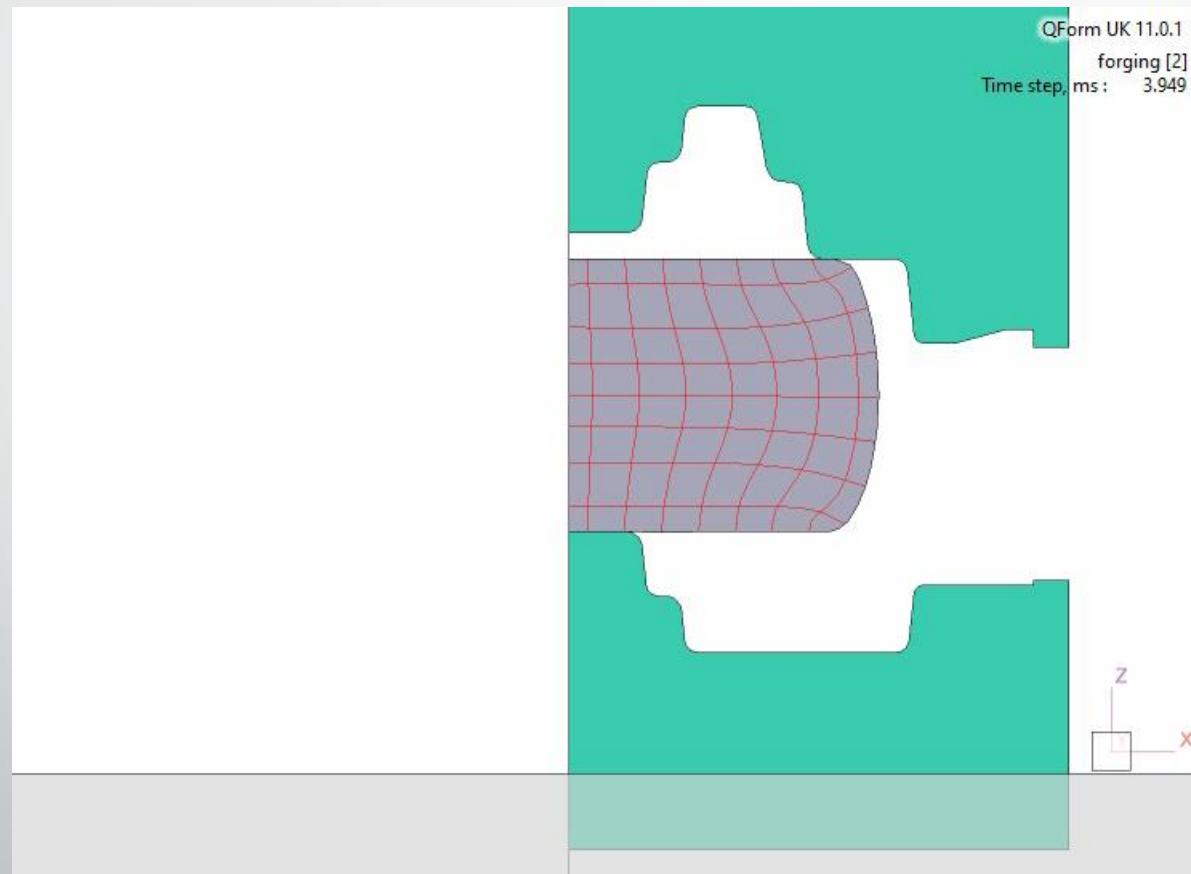
Upsetting



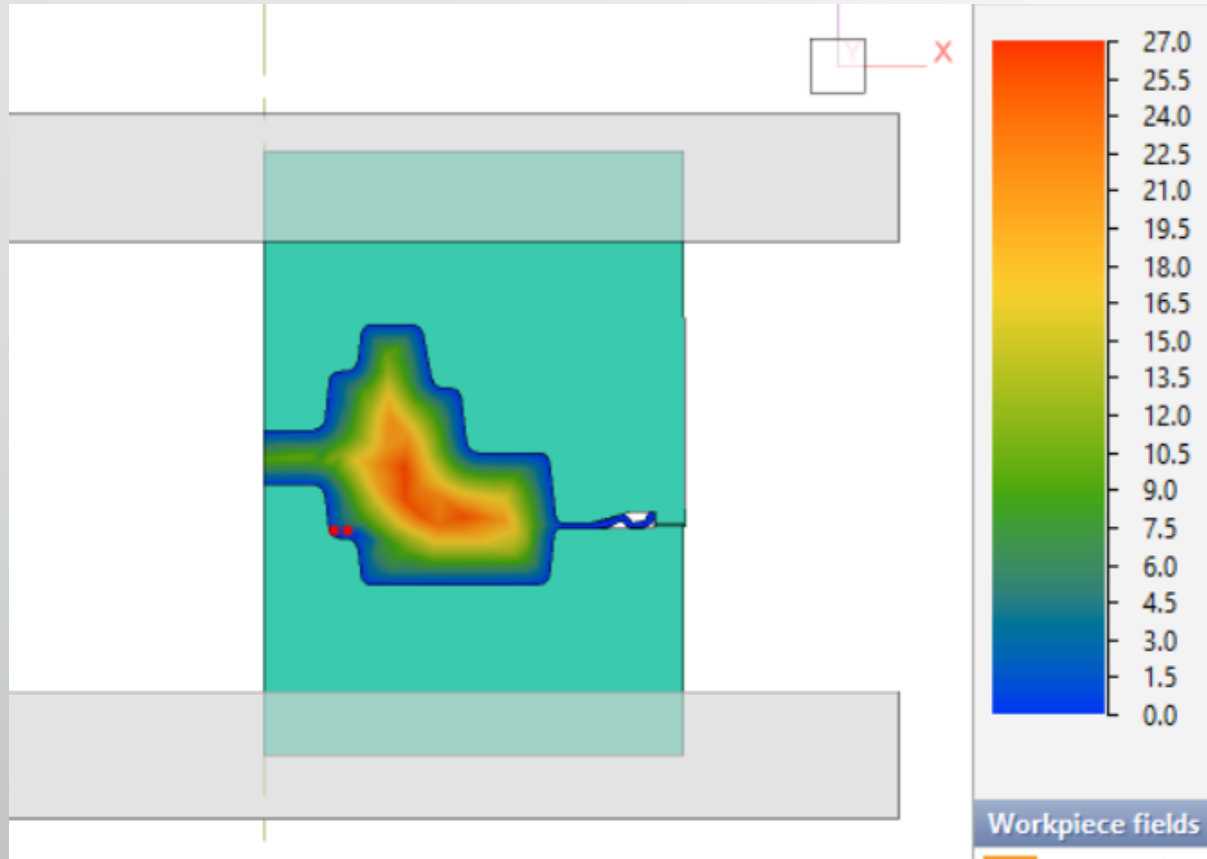
Final Assembly before the simulation



2D Q form simulation



Q-Form Simulations



After performing the simulation we can see that there are some defects present in the workpiece because the diameter was not optimal, also there was problem with the flash part.(only present in 2D)

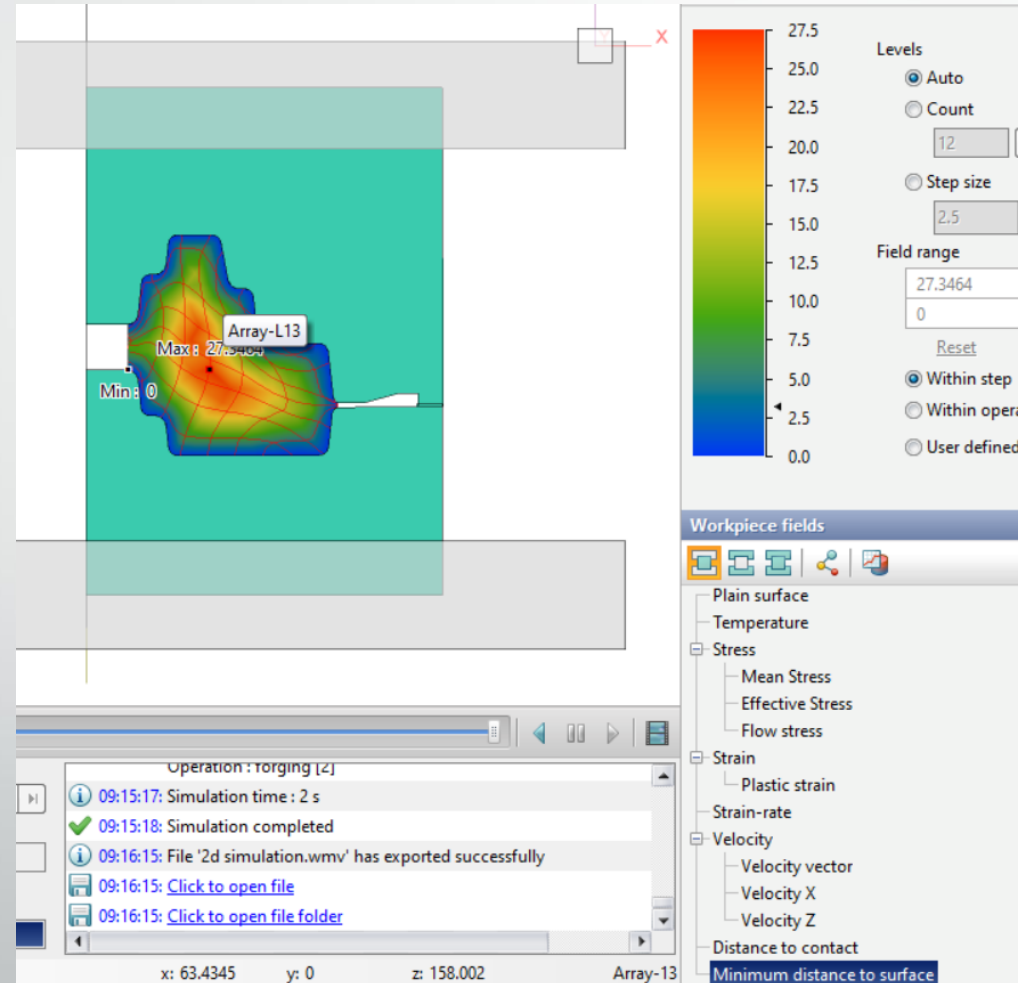
The problem with the flashed part will be solved by trimming.

Updating the billet diameter

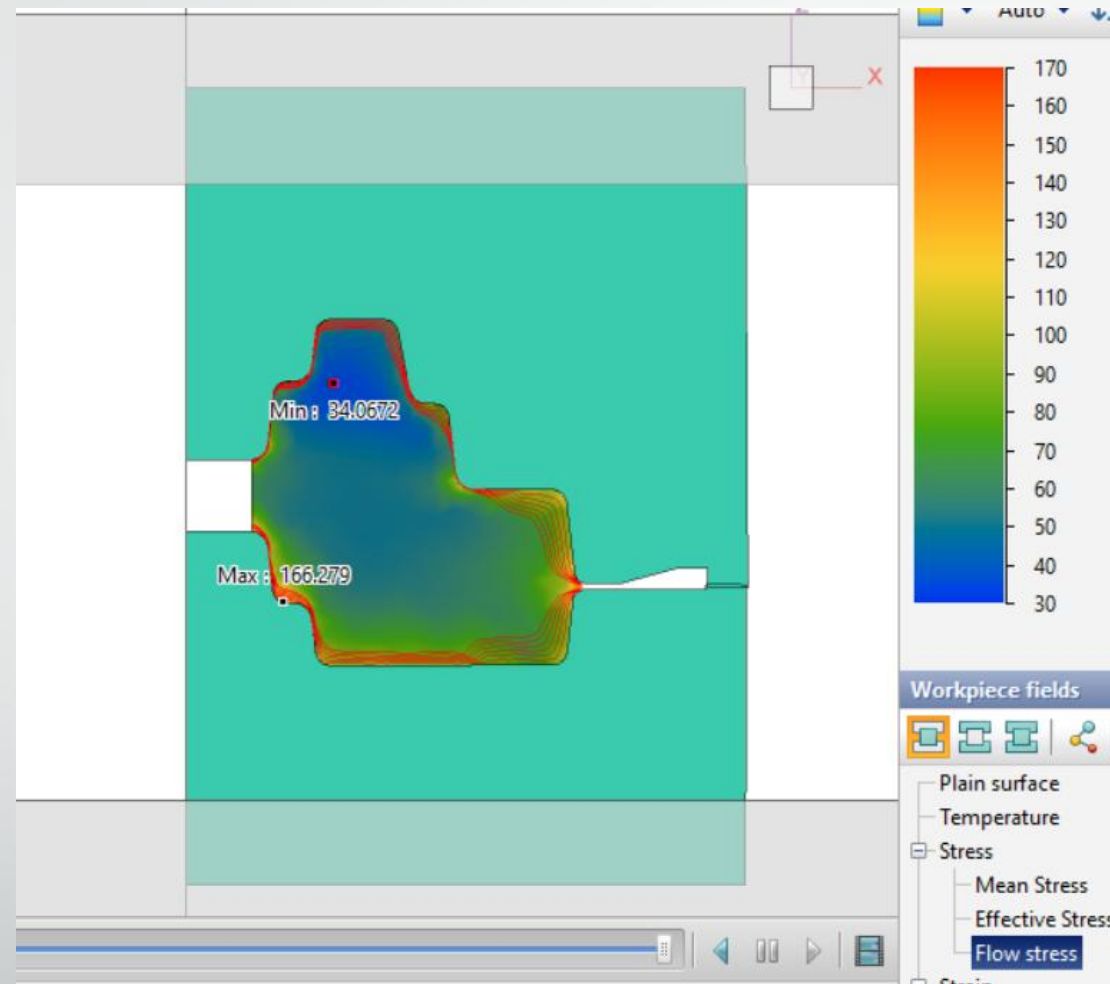
Material density (kg/mm ³):	0.000007514
Category of shape:	2
K:	1.08
Mass forged part * K:	20.736
Total volume:	2759648.66
Diameter Dg (mm):	140
Height Hg (mm):	180.00
$h/d < 2,5$	1.29

We decreased the height so the diameter is higher and tried the simulation again.

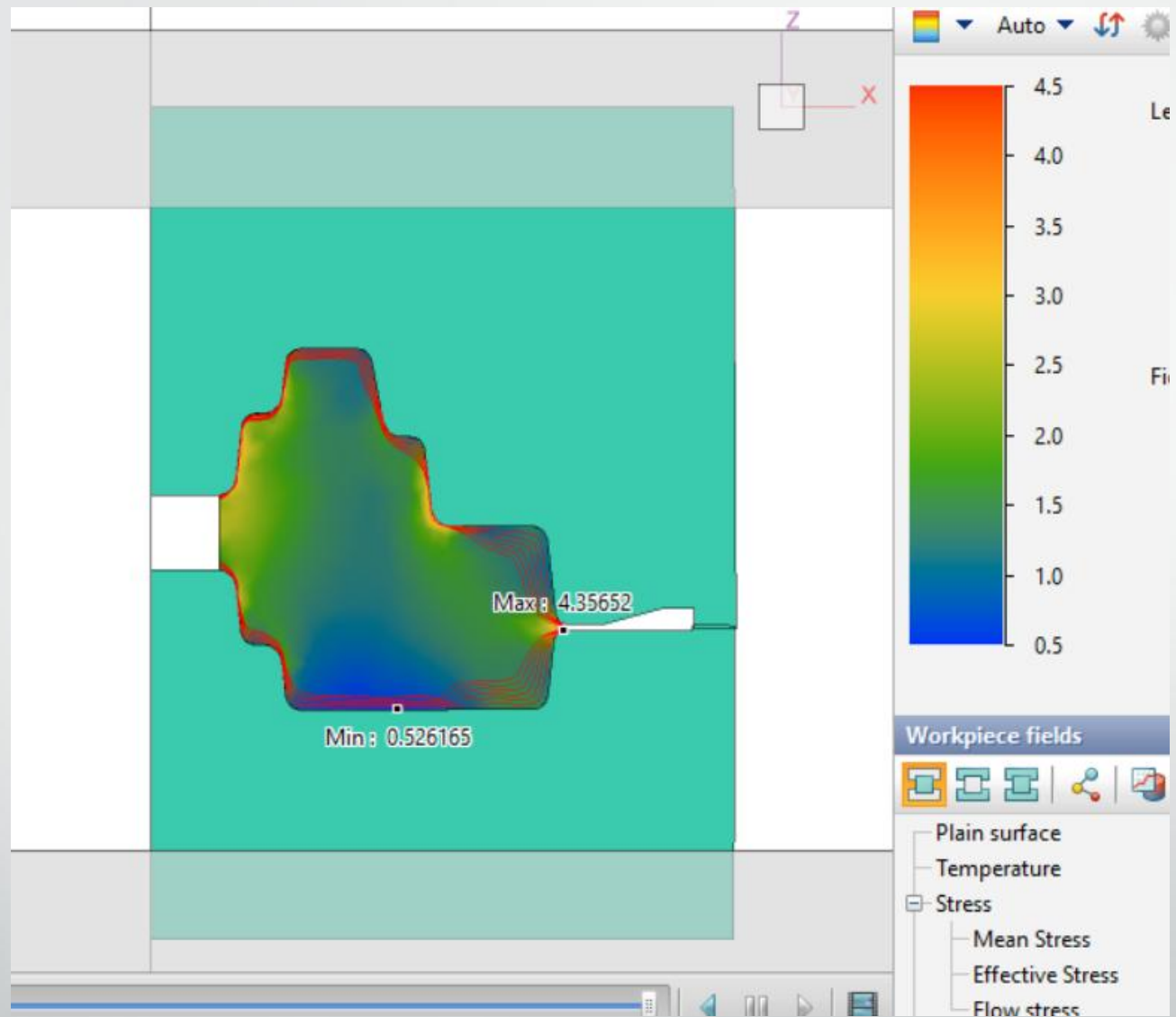
2D q form simulation after updates (dimensions and trimming)



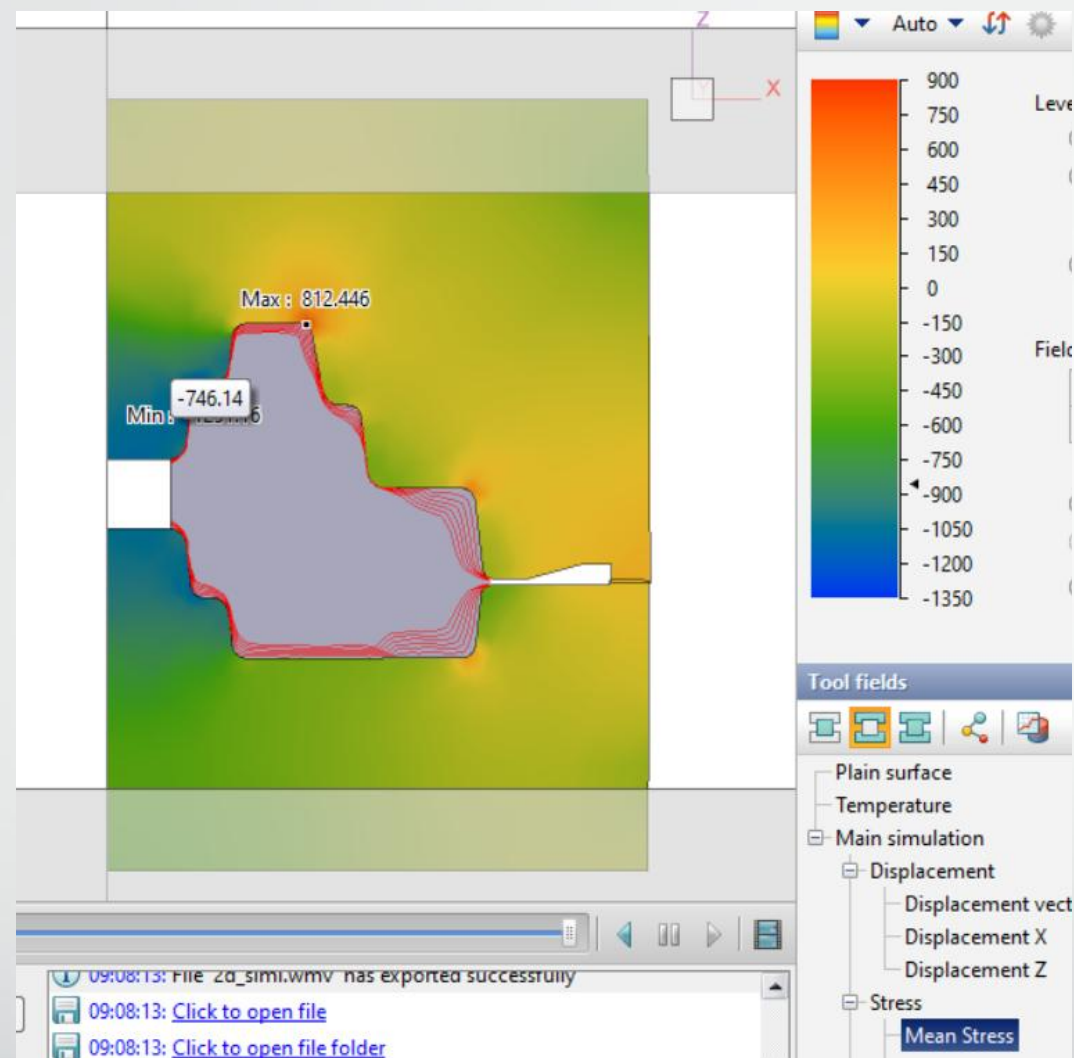
Some results from the 2d simulation



Maximum and minimum stress in the workpiece

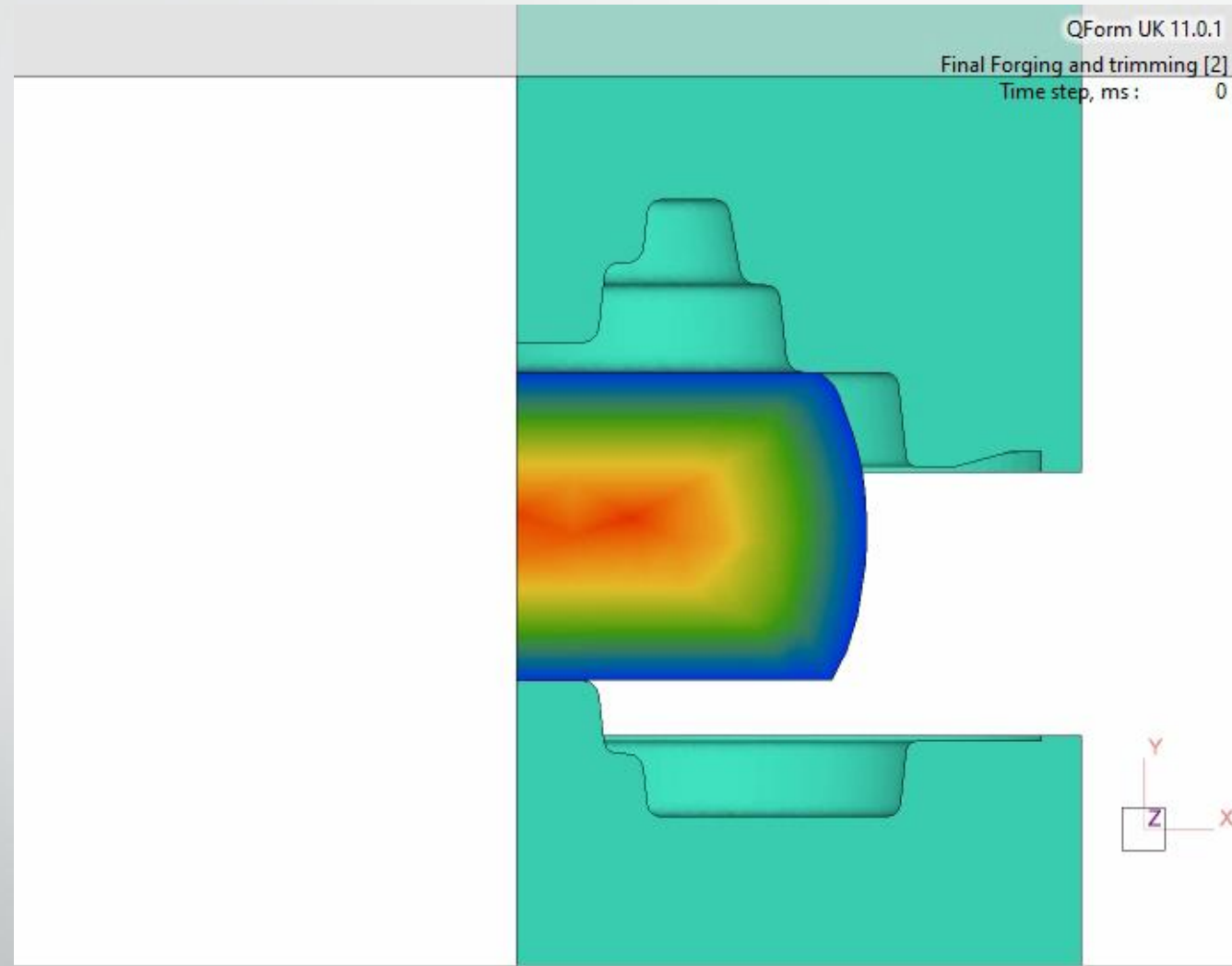


Maximum and minimum strain in the workpiece

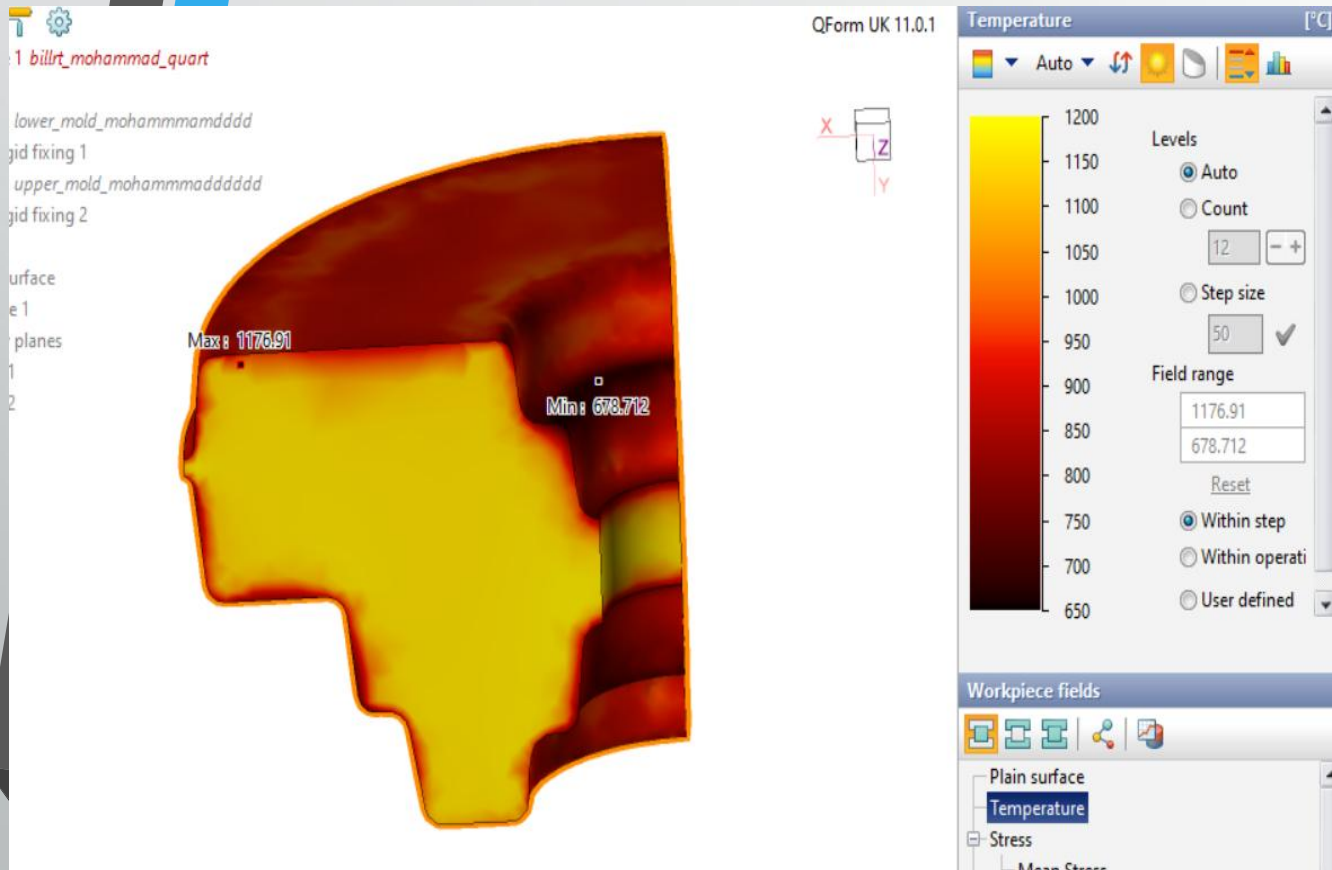


Maximum and minimum mean(principal stress in the dice

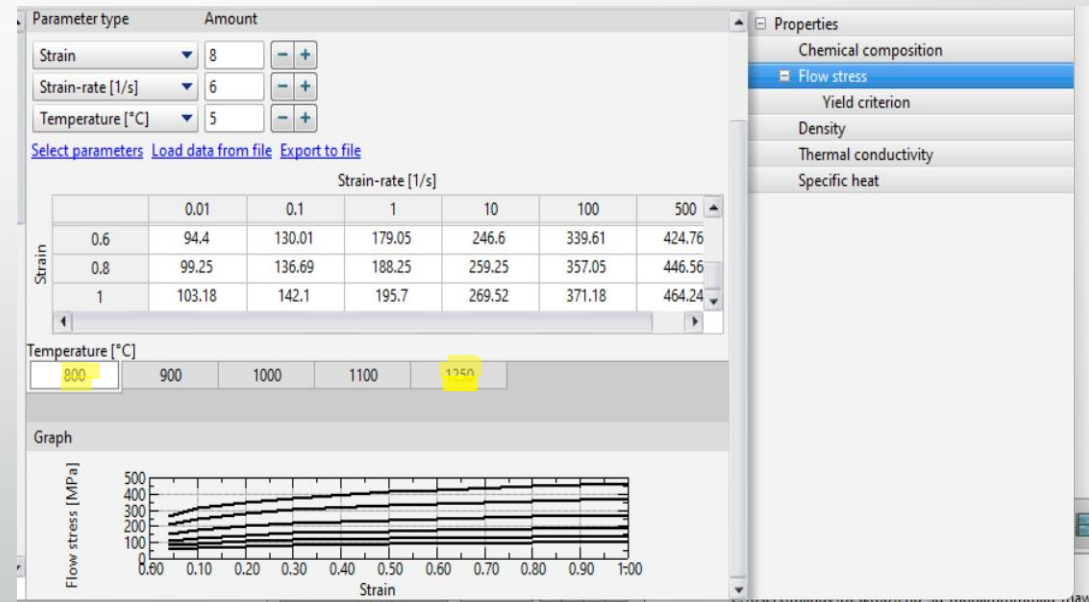
3D q form simulation



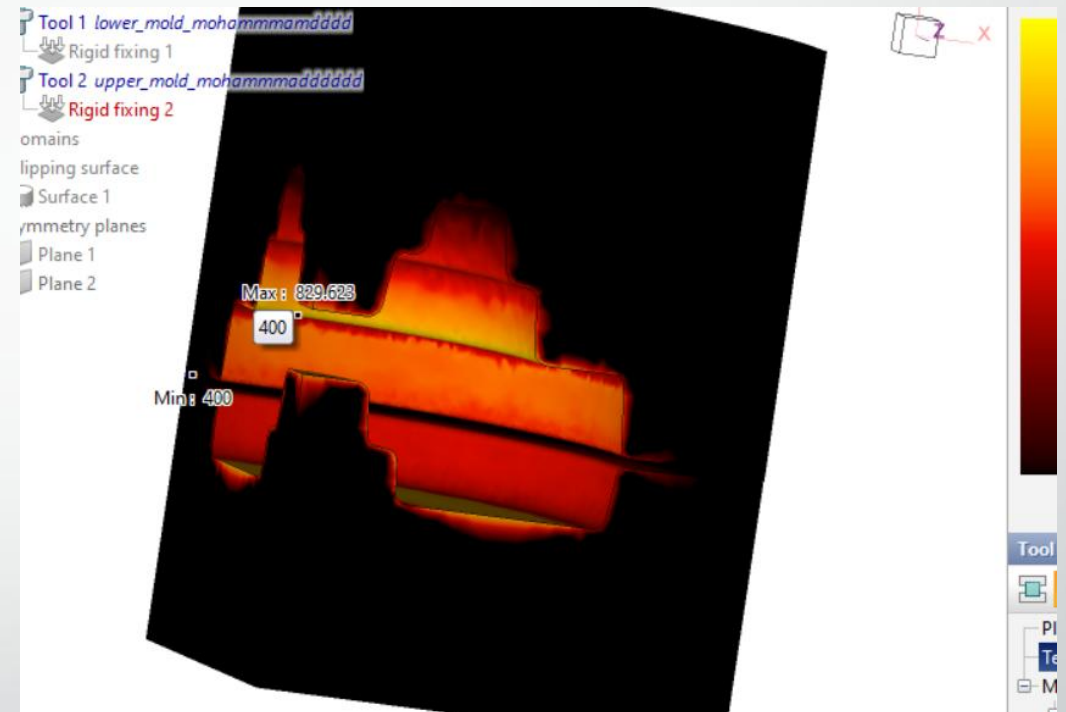
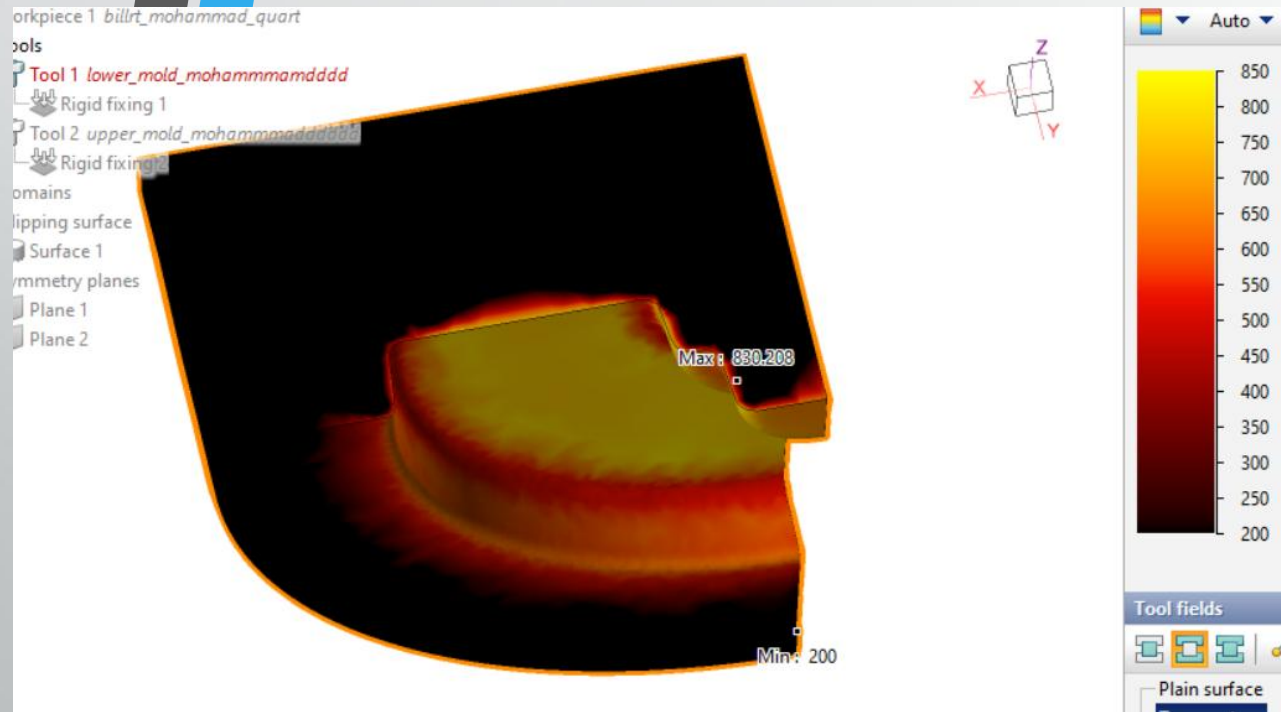
Results of workpiece simulation



We can see that the minimum temperature is 678,712 which is low in compared to the acceptable range (800-1250)

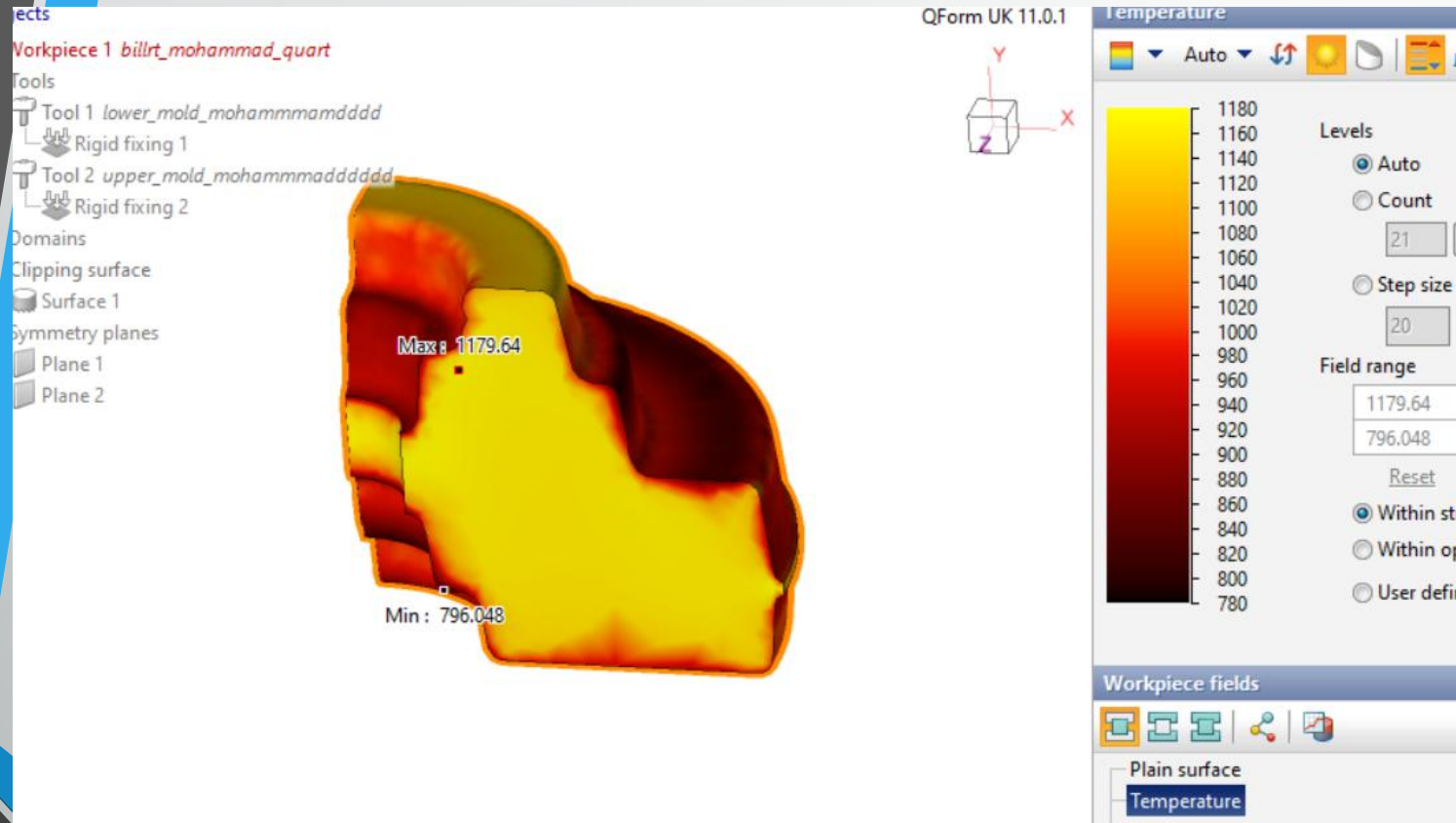


Tool Temperature Results before changing the lubricant



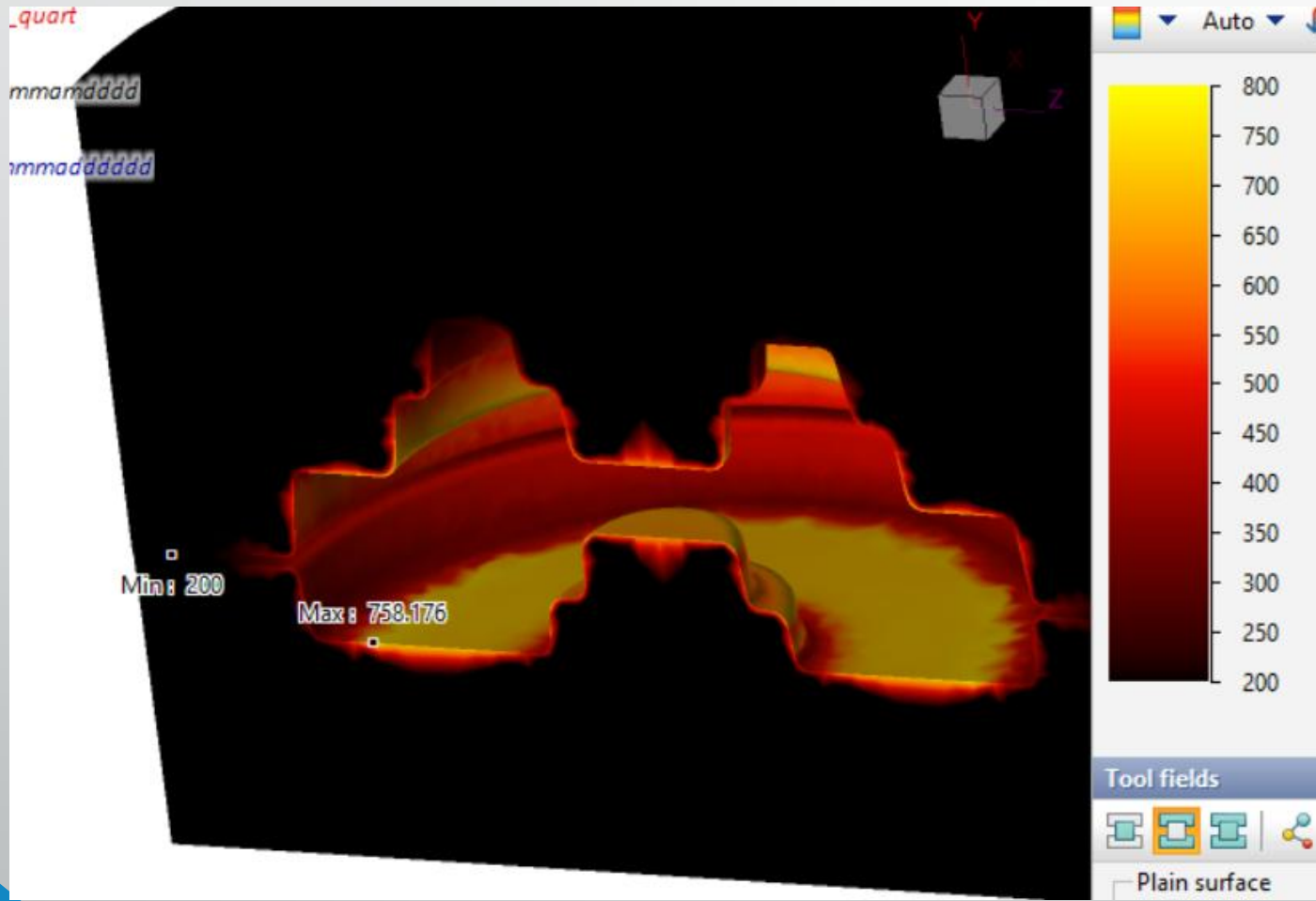
The temperature of the tools are not in the acceptable range, the temperature be lower because it is a continuous process so the temperature can damage it.

After chaging the lubricant from Graphite + water to GLASS



Now the minimum temperature of the workpiece has increased after changing the lubricant to min 800

Tools Temperature after changing the lubricant



After changing the lubricant we can see a decrease in the temperature of the tools

Glass lubricant decreases the friction so the dice will not increase their temperature and workpiece will stay the same.



Thank you!